

Architectural Design Document: Network-Aware Mine Haulage Prediction

Project: GNN-Based Mine Haulage Prediction with Failure Awareness and Uncertainty

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1 Executive Summary & Problem Formulation

In open-pit mine haulage operations, predicting cycle times accurately across a road network is critical for automated truck dispatching and short-term scheduling. Traditional mining operations rely on flat, tabular machine learning models (e.g., XGBoost) or static empirical physics models. However, haulage networks operate as dynamic, directed graphs where equipment failures, queue buildups, and localized congestion propagate spatially across road segments. This design document establishes the empirical limitations of tabular modeling based on our Week 1 benchmark findings on 4,800 historical haulage observations and justifies the progressive transition from a **Graph Convolutional Network (GCN)** to a **Graph Attention Network (GAT)** integrated with Deep Ensemble uncertainty quantification.

2 Empirical Justification: The Tabular Blind Spot (Week 1 Findings)

Our Week 1 benchmark evaluation of a tabular XGBoost regressor achieved a **Loaded Trips MAE of 85.76 s** (beating the syllabus estimated baseline of ~ 101 s) and a **Combined Trips MAE of 78.72 s**. While strong, an analysis of the model’s feature importance hierarchy revealed a critical empirical limitation:

Feature Category	Primary Variables	Importance	Operational Interpretation
Deterministic Physics	loaded, max_gradient_pct	65.89%	Tabular trees excel at static vehicle mechanics (engine power vs. gravity).
Global State Flag	failure_active	25.16%	Detects <i>when</i> a failure occurs, but applies a spatially invariant penalty.
Network Congestion	congestion_level, payload_t	3.36%	The Tabular Blind Spot: Fails to utilize localized queue states.

The Architectural Flaw: Why did `congestion_level` account for just 3.36% of model importance when physical traffic is the primary operational cause of mine haulage delays? Because a tabular tree treats observations as independent, identically distributed (*i.i.d.*) rows. When an equipment failure (`failure_active = 1`) occurs at Crusher CR, XGBoost cannot trace the directed road segments to determine *which* approaching trucks are trapped in the resulting cascade vs. which trucks are operating on uncongested waste dumps across the pit. To close the remaining **5.76 s error gap** and hit our **<80.00 s Loaded MAE KPI target**, we must transition to an architecture natively capable of spatial message passing across a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

3 Why Graph Convolutional Networks (GCNs) Are Insufficient

For our Week 2 baseline, we implement a 2-layer Graph Convolutional Network (Kipf & Welling, 2017). A standard GCN updates node feature representations $\mathbf{H}$ using the isotropic propagation rule:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right) \quad (1)$$

where $\tilde{\mathbf{A}}$ is the adjacency matrix with added self-loops and $\tilde{\mathbf{D}}$ is the diagonal degree matrix.

The Limitation for Mining Operations: Standard GCNs utilize **symmetric, isotropic aggregation** ($\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$). This mathematical formulation forces a node to aggregate information from all adjacent neighbors with static weights dictated purely by topological degree. In a mine haulage network, this is physically flawed: an intersection connected to two empty waste dumps and one bottlenecked, failing ore shovel will weigh messages from all three destinations roughly equally. Standard GCNs lack the capacity to dynamically prioritize active operational bottlenecks over passive, uncongested road corridors.

4 Selected Architecture: Graph Attention Networks (GAT)

To overcome isotropic aggregation, our primary predictive engine will utilize a **Graph Attention Network (GAT)** (Veličković et al., 2018). GATs replace fixed degree-normalization with learnable, anisotropic attention coefficients α_{ij} , which compute the relative importance of neighbor j ’s features onto node i :

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))} \quad (2)$$

Why GAT is the Optimal Architecture for Mine Haulage:

1. **Dynamic Bottleneck Routing:** The attention mechanism allows the network to dynamically learn that when a shovel’s queue feature (`queue_capacity` / queue state) spikes, adjacent approaching road arcs must heavily attend ($\alpha_{ij} \rightarrow 1$) to that node’s state, while ignoring uncongested diverging paths ($\alpha_{ik} \rightarrow 0$).
2. **Failure-Aware Propagation (Week 4 Integration):** When a crusher undergoes an unplanned failure (zero capacity state), GAT layers instantly shift attention weights across the graph, mathematically modeling how traffic rerouting impacts travel times on adjacent corridors without requiring manual topological rule-coding.
3. **Operational Interpretability (Week 3 Deliverable):** By extracting and mapping the top-3 attention weights per arc onto a spatial mine map, we provide dispatch supervisors with an interpretable visual heatmap proving *why* the model predicted a delay on a specific road segment.

5 Uncertainty Quantification & Downstream Optimizer Integration

To support the downstream Two-Stage Stochastic MILP truck-dispatch optimizer (Week 7), point-predictions of travel time are insufficient; the optimizer requires calibrated probability distributions to manage risk.

We implement a **Deep Ensemble of $M = 10$ independently seeded GAT networks**. For any directed haulage arc, the ensemble mean μ serves as the expected cycle time, while the sample variance σ^2 captures epistemic and aleatoric uncertainty:

$$\mu = \frac{1}{M} \sum_{m=1}^M \hat{y}_m, \quad \sigma^2 = \frac{1}{M-1} \sum_{m=1}^M (\hat{y}_m - \mu)^2 \quad (3)$$

This explicit variance parameter (σ^2) directly informs the Sample Average Approximation (SAA) scenarios within the MILP optimizer. When a road corridor exhibits high predictive variance (e.g., during active equipment failures or severe shift changes), the MILP automatically widens its safety margins or routes trucks to alternative, lower-variance paths—proving that uncertainty-aware prediction improves operational dispatch decisions.